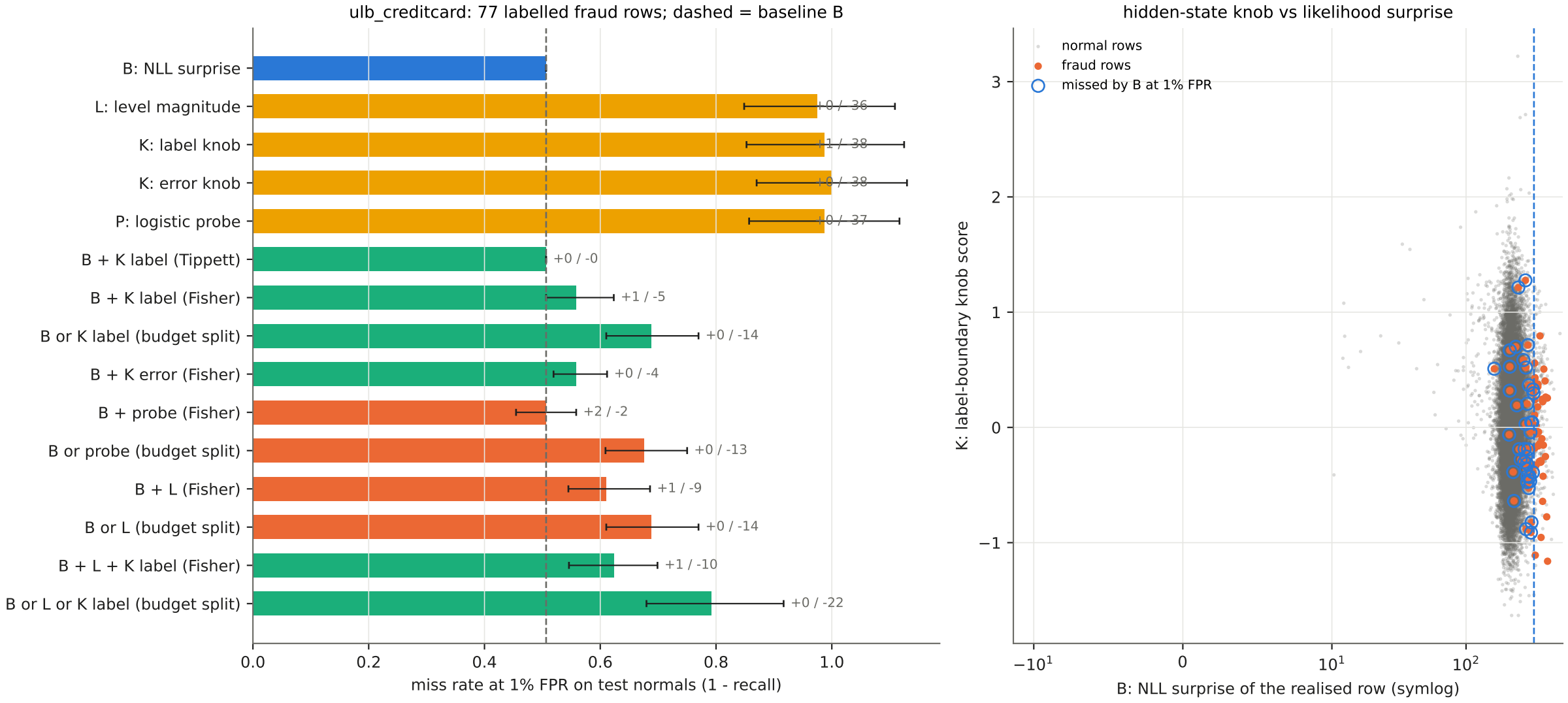

ulb_creditcard: 77 labelled fraud rows; dashed = baseline B
B: NLL surprise
L: level magnitude
K: label knob
K: error knob
P: logistic probe
B + K label (Tippett)
B + K label (Fisher)
B or K label (budget split)
B + K error (Fisher)
B + probe (Fisher)
B or probe (budget split)
B + L (Fisher)
B or L (budget split)
B + L + K label (Fisher)
B or L or K label (budget split)
+0 / -36
+1 / -38
+0 / -38
+0 / -37
+0 / -0
+1 / -5
+0 / -14
+0 / -4
+2 / -2
+0 / -13
+1 / -9
+0 / -14
+1 / -10
+0 / -22
0.0
0.2
0.4
0.6
0.8
1.0
miss rate at 1% FPR on test normals (1 - recall)
hidden-state knob vs likelihood surprise
normal rows
fraud rows
missed by B at 1% FPR
K: label-boundary knob score
3
2
1
0
−1
−10^1
0
10^1
10^2
B: NLL surprise of the realised row (symlog)